Four points on the unit circle

Putnam, 2020 B5

For $j \in \{1, 2, 3, 4\}$, let z_j be a complex number with $|z_j| = 1$ and $z_j \neq 1$. Prove that

$$3 - z_1 - z_2 - z_3 - z_4 + z_1 z_2 z_3 z_4 \neq 0.$$

Proof. Suppose, to the contrary, that

$$3 - z_1 - z_2 - z_3 - z_4 + z_1 z_2 z_3 z_4 = 0. \tag{1}$$

We first observe that if z is a complex number with $|z| = 1$ and $z \neq 1$, then

$$\operatorname{Re}\big((1 - z)^{-1}\big) = \frac{1}{2}.$$

Indeed, write $a = \operatorname{Re} z$ and $b = \operatorname{Im} z$. Since $|z| = 1$, we have

$$a^2 + b^2 = 1.$$

Also $1 - z \neq 0$, and

$$|1 - z|^2 = (1 - a)^2 + b^2 = 1 - 2a + a^2 + b^2 = 2 - 2a,$$

which is nonzero. Since $(1 - z)^{-1} = \overline{1 - z}/|1 - z|^2$, it follows that

$$\operatorname{Re}\big((1 - z)^{-1}\big) = \frac{1 - a}{|1 - z|^2} = \frac{1 - a}{2 - 2a} = \frac{1}{2}.$$

Since $z_j \neq 1$, we have $1 - z_j \neq 0$ for each j. Put

$$w_j = (1 - z_j)^{-1} \qquad (j = 1, 2, 3, 4),$$

so that, by the observation above,

$$\operatorname{Re} w_j = \frac{1}{2} \qquad (j = 1, 2, 3, 4).$$

Next, we have the identity

$$\begin{aligned} &3 - z_1 - z_2 - z_3 - z_4 + z_1 z_2 z_3 z_4 \\ &\quad = \big((1 - z_1)(1 - z_2)(1 - z_3)(1 - z_4)\big) \\ &\qquad \cdot \Big(1 - w_1 - w_2 - w_3 - w_4 + w_1 w_2 + w_1 w_3 + w_1 w_4 + w_2 w_3 + w_2 w_4 + w_3 w_4\Big). \end{aligned}$$

To see this, set $p_j = 1 - z_j$, so $z_j = 1 - p_j$. Expanding $3 - \sum_j z_j + \prod_j z_j$ in terms of the p_j gives the sum of the pairwise products $p_j p_k$ minus the sum of the triple products $p_j p_k p_\ell$ plus $p_1 p_2 p_3 p_4$, which is exactly what is obtained by multiplying out the right-hand side, since $w_j = p_j^{-1}$.

By assumption (1), the left-hand side is 0. Since each $1 - z_j$ is nonzero, the product $(1-z_1)(1-z_2)(1-z_3)(1-z_4)$ is nonzero. Hence the second factor must vanish:

$$1 - w_1 - w_2 - w_3 - w_4 + w_1w_2 + w_1w_3 + w_1w_4 + w_2w_3 + w_2w_4 + w_3w_4 = 0. \quad (2)$$

Now write

$$w_j = \frac{1}{2} + it_j, \qquad \text{where } t_j = \operatorname{Im} w_j \in \mathbb{R}.$$

For $j < k$,

$$w_j w_k = \left(\frac{1}{4} - t_j t_k\right) + i\,\frac{t_j + t_k}{2}.$$

Summing over the six pairs $j < k$, and noting that each t_j occurs in exactly three pairs, we obtain

$$\sum_{1\le j<k\le 4} w_j w_k = \left(\frac{3}{2} - \sum_{1\le j<k\le 4} t_j t_k\right) + \frac{3i}{2}(t_1 + t_2 + t_3 + t_4),$$

while

$$w_1 + w_2 + w_3 + w_4 = 2 + i(t_1 + t_2 + t_3 + t_4).$$

Therefore the left-hand side of (2) equals

$$\left(\frac{1}{2} - \sum_{1\le j<k\le 4} t_j t_k\right) + \frac{i}{2}(t_1 + t_2 + t_3 + t_4).$$

Since this is 0, both its real and imaginary parts are 0. Thus

$$\frac{t_1 + t_2 + t_3 + t_4}{2} = 0$$

and

$$\frac{1}{2} - \big(t_1t_2 + t_1t_3 + t_1t_4 + t_2t_3 + t_2t_4 + t_3t_4\big) = 0.$$

In other words,

$$t_1 + t_2 + t_3 + t_4 = 0 \quad (3)$$

and

$$t_1t_2 + t_1t_3 + t_1t_4 + t_2t_3 + t_2t_4 + t_3t_4 = \frac{1}{2}. \quad (4)$$

But expanding the square gives

$$(t_1 + t_2 + t_3 + t_4)^2 = (t_1^2 + t_2^2 + t_3^2 + t_4^2) + 2\big(t_1t_2 + t_1t_3 + t_1t_4 + t_2t_3 + t_2t_4 + t_3t_4\big).$$

Using (3) and (4), this becomes

$$0 = (t_1^2 + t_2^2 + t_3^2 + t_4^2) + 1,$$

so

$$t_1^2 + t_2^2 + t_3^2 + t_4^2 = -1.$$

This is impossible, since $t_1^2 + t_2^2 + t_3^2 + t_4^2 \geq 0$ for real t_1, t_2, t_3, t_4. This contradiction shows that (1) cannot hold, and hence

$$3 - z_1 - z_2 - z_3 - z_4 + z_1 z_2 z_3 z_4 \neq 0,$$

as required. □